

Load Interconnection Request System Impact Study Report

For Queue Project Q23-2294

Clarence 100 kV
March 22, 2025

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q23-2294. The Interconnection Customer proposes a 127.0 MW large load (data center) facility interconnecting at the Clarence 100 kV substation in the Ohio area. The request was received on March 10, 2023. The study evaluated system intact and single contingency conditions using the regional planning case, including 428 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 17 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$64,948,000.

3. Network Impacts

3.1 Load Deliverability Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
West Olympia-Harper 100.0 kV Cir. 1	W Olympia-Harper 100.0 kV Line	174.2	190	5.9
Harper-West Olympia Substation 100.0 kV Line 2	W Olympia-Harper 100.0 kV	174.2	190	5.9
R.P.Mone-Monroeville 100.0 kV	Dekalb-Jones Chapel 500.0 kV Line	142.2	104	6.7
Pruitt Escalante Pegs-Clarence 100.0 kV	Swan Vly-Muskogee 500.0 kV	140.5	151	57.5
Taconite Harbor Energy Center-Pruitt Escalante Pegs 100.0 kV	Swan Valley Substation-Muskogee Substation 500.0 kV	136.0	167	57.5
Dixon-Muskogee 100.0 kV	Swan Valley-Muskogee Substation 500.0 kV	127.7	148	26.8
Ephraim Cemetery-Taiban Mesa 100.0 kV	Taconite Harbor Energy Ctr Substation-Pruitt Escalante Pegs Substation 100.0 kV	126.7	129	100.0
Taconite Harbor Energy Center-Fort Thomson Substation 100.0 kV	Swan Vly-Muskogee 500.0 kV	122.6	143	26.8
Dixon-Ft Thompson 100.0 kV L	Muskogee-Swn Valley 500.0 kV	121.4	142	26.8
Madelia-R.P.Mone 100.0 kV	Jones Chapel-Dekalb 500.0 kV Line	118.1	117	6.7
N. Lee-Muskogee 100.0 kV	Dixon-Muskogee 100.0 kV	117.4	143	38.5
Clarence-Ephraim Cemetery 100.0 kV	Pruitt Escalante Pegs-Taconite Harbor Energy Ctr 100.0 kV Line	114.9	136	100.0
N. Lee-Taconite Harbor Energy Ctr 100.0 kV	Muskogee-Dixon 100.0 kV L	114.0	137	38.5

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Bluebonnet-Jones Chapel 230.0 kV	Robison Prk-Russell Fork 230.0 kV	113.2	1,701	6.1
Barret-Roseau County 100.0 kV	Harper-Monte Del Estado 6303 100.0 kV	106.9	364	8.0
Pleasant Street-Lake Marion 100.0 kV	Bark River-San Andres 100.0 kV	104.7	112	6.9
Chimacum-Madelia 100.0 kV L	Dekalb-Jones Chapel 500.0 kV	100.4	125	6.7

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
W Olympia-Harper 100.0 kV Circuit 1	Rebuild line with higher-capacity conductor	\$2,314,000	56% (\$1,296,000)
Harper-West Olympia Substation 100.0 kV Line 2	Rebuild line with higher-capacity conductor	\$2,644,000	56% (\$1,481,000)
R.P.Mone-Monroeville 100.0 kV	Rebuild line with higher-capacity conductor	\$5,213,000	65% (\$3,400,000)
Pruitt Escalante Pegs-Clarence 100.0 kV	Rebuild line with higher-capacity conductor	\$4,087,000	100% (\$4,087,000)
Taconite Harbor Energy Ctr-Pruitt Escalante Pegs Substation 100.0 kV	Rebuild line with higher-capacity conductor	\$8,474,000	100% (\$8,474,000)
Dixon-Muskogee 100.0 kV	Rebuild line with higher-capacity conductor	\$2,850,000	100% (\$2,850,000)
Ephraim Cemetery-Taiban Mesa 100.0 kV	Rebuild line with higher-capacity conductor	\$6,535,000	100% (\$6,535,000)
Taconite Harbor Energy Center-Fort Thomson Substation 100.0 kV	Rebuild line with higher-capacity conductor	\$2,475,000	100% (\$2,475,000)
Dixon-Ft Thompson 100.0 kV L	Rebuild line with higher-capacity conductor	\$1,406,000	100% (\$1,406,000)
Madelia-R.P.Mone 100.0 kV	Rebuild line with higher-capacity conductor	\$8,951,000	65% (\$5,837,000)
N. Lee-Muskogee 100.0 kV	Rebuild line with higher-capacity conductor	\$6,949,000	90% (\$6,288,000)
Clarence-Ephraim Cemetery 100.0 kV	Rebuild line with higher-capacity conductor	\$11,721,000	100% (\$11,721,000)
N. Lee-Taconite Harbor Energy Ctr 100.0 kV	Rebuild line with higher-capacity conductor	\$2,295,000	90% (\$2,077,000)
Bluebonnet-Jones Chapel 230.0 kV	Rebuild line with higher-capacity conductor	\$47,143,000	0% (\$150,000)
Barret-Roseau County 100.0 kV	Reconductor line	\$1,919,000	44% (\$843,000)
Pleasant Street-Lake Marion 100.0 kV	Reconductor line	\$4,907,000	100% (\$4,907,000)
Madelia-Chimacum 100.0 kV	Reconductor line	\$1,719,000	65% (\$1,121,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
	Total allocated to project		\$64,948,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.